

Strategic Credit Risk Masterclass

Predictive Power and Mathematical Interpretability

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The Necessity of Interpretability

- **Regulatory Compliance:** Credit decisions must be explainable to regulators and consumers.
- **Reason Codes:** Black-box models are unsuitable because they cannot provide clear reasons for rejection.
- **Logistic Regression:** Remains the industry standard due to its linear, transparent nature.

The Modelling Challenge: How to capture non-linear risk patterns within a strictly linear constraint?

Weight of Evidence (WoE) Theory

Weight of Evidence transforms raw features into a risk-based scale by comparing the distribution of Good vs. Bad customers in specific buckets.

The Mathematical Definition

$$WoE_i = \ln \left(\frac{\% \text{ Good Customers}_i}{\% \text{ Bad Customers}_i} \right)$$

- **Positive WoE:** Higher proportion of Goods (Safe bin).
- **Negative WoE:** Higher proportion of Bads (Risky bin).
- **Zero WoE:** Neutral risk (Portfolio average).

Information Value (IV): Quantifying Signal

IV measures the total predictive power of a feature across all bins. It helps rank variables for feature selection.

IV Formula

$$IV = \sum_{i=1}^n (\% \text{ Goods}_i - \% \text{ Bads}_i) \times WoE_i$$

IV Range	Predictive Power
≤ 0.02	Not useful
0.02 – 0.1	Weak
0.1 – 0.3	Medium
0.3 – 0.5	Strong

Scorecard Scaling: From Probabilities to Points

We use **Points to Double Odds (PDO)** to scale logistic log-odds into a user-friendly score (e.g., 300–900).

- **Factor:** Determines the weight of score increase per unit of risk.

$$Factor = \frac{PDO}{\ln(2)}$$

- **Offset:** Aligns the score to a target anchor (e.g., 600 points).

$$Offset = \text{Target Score} - (Factor \times \ln(\text{Target Odds}))$$

The Mathematical Core: Intercept Distribution

To ensure the sum of points equals the total score, the model intercept (α) must be distributed across all n features.

Point Calculation Formula

$$Points_{ij} = -(\beta_i \times WoE_{ij} \times Factor) - \left(\frac{\alpha}{n} \times Factor \right) + \frac{Offset}{n}$$

This architectural step ensures that the baseline risk of the portfolio is accurately captured in every single scorecard attribute.

Population Stability Index (PSI)

PSI monitors if the characteristics of incoming applicants have shifted away from the training population.

PSI Formula

$$PSI = \sum (Actual\% - Expected\%) \times \ln \left(\frac{Actual\%}{Expected\%} \right)$$

- **PSI $\leq$ 0.1:** Stable population.
- **PSI $>$ 0.2:** Major data drift; model requires retraining.